

Analytical software developer with a strong background in machine learning, data science and full-stack development. I am dedicated to solving complex problems and am pursuing opportunities to contribute my skills to a team focused on creating high-quality, innovative products.

Experience

AI Enablement Intern

February 2026 – Currently

Micron

Guadalajara, México

- Designed and implemented a Python and Rasa framework-based Chatbot to facilitate new developer onboarding and provide troubleshooting assistance, complete with a local SQL database for response management, a ChatGPT-based model API for enhanced query handling, and Github for version control.
- Automating report generation for fault-mapping between customer defined fault application and ECU's real monitors using Python scripts, leveraging Pandas, Anaconda, and fuzzy logic libraries, to ensure accurate tracking of monitor-fault mappings and to guarantee software quality deliverables.
- Developed scripts to cross-reference client requirements with ECU header files, identifying potential inconsistencies and streamlining the verification process.

Key Metrics: Enhanced quality of deliverables and improved efficiency by 40%, reducing inconsistencies by 90%.

Software Engineer Intern

August 2023 – July 2024

Robert Bosch México

Guadalajara, México (Hybrid)

- Designed and implemented a Python and Rasa framework-based Chatbot to facilitate new developer onboarding and provide troubleshooting assistance, complete with a local SQL database for response management, a ChatGPT-based model API for enhanced query handling, and Github for version control.
- Automating report generation for fault-mapping between customer defined fault application and ECU's real monitors using Python scripts, leveraging Pandas, Anaconda, and fuzzy logic libraries, to ensure accurate tracking of monitor-fault mappings and to guarantee software quality deliverables.
- Developed scripts to cross-reference client requirements with ECU header files, identifying potential inconsistencies and streamlining the verification process.

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Education

University of Guadalajara

Guadalajara, México

Master's Degree in Machine Learning and Artificial Intelligence

January 2025 – Currently

- Implementing and optimizing custom LSTM architectures for time-series forecasting using PyTorch and Python. Focus on improving computational efficiency and long-term dependency retention.

University of Guadalajara

Guadalajara, México

Bachelor's Degree in Robotics Engineering

January 2020 – June 2024

- Bachelor's degree providing hands-on training in developing machine learning algorithms and robotic vision systems (Python, Matlab), designing and programming the underlying electronic circuits and embedded system (C/C++ and VHDL).

Projects

Automated MLOps System V2 | [Live](#) | [Github](#) | [API](#)

Python, FastAPI, Docker, GitHub Actions, Prefect, OCI

- Engineered a self-healing MLOps pipeline that automates the model lifecycle for shelf-life prediction, utilizing GitHub Actions for CI/CD to build and deploy Docker containers upon code changes.
- Implemented a Hybrid Cloud architecture, orchestrating training jobs on Kaggle GPUs while centralizing artifact tracking on a custom MLflow server deployed on OCI.
- Developed a REST API with FastAPI that supports hot-reloading of models, automatically promoting superior models based on live metric evaluation.

Modular AI Interface Library | [Live](#) | [Github](#)

React, Next.js, Express, Tailwind CSS

- Architected a distributable component library (lep-chat) using Rollup to bundle optimized ESM/CommonJS builds with full TypeScript definitions.
- Developed a production-ready Full-Stack demo using Next.js 15, implementing secure server-side API routes to orchestrate communication between the frontend and Gemini LLM.
- Engineered a zero leakage styling system combining Tailwind CSS and CSS Modules to ensure isolation.

Technical Skills

Programming Languages: Python, C/C++, Javascript, Typescript, HTML, CSS

Technologies: Docker, CI/CD, Linux, Tensorflow, PyTorch, Sckit-learn, Oracle Cloud, Prefect, FastAPI, Pandas, React JS, Next JS, Node, Express JS, Tailwind, SQLite, PostgreSQL, Supabase, Excel, PowerBI

Tools: Git, GitHub, Visual Studio Code

Certifications

- **Oracle Cloud Infrastructure 2025 AI Foundations Associate (1Z0-1122-25)** | *Issued: Aug 2025* | [Verify](#)